
3.4.5 Energy is transferred through ecosystems and the efficiency of transfer can be measured.

Energy transfer

Photosynthesis is the main route by which energy enters an ecosystem.

Energy is transferred through the trophic levels in food chains and food webs and is dissipated.

Quantitative consideration of the efficiency of energy transfer between trophic levels.

Pyramids of numbers, biomass and energy and their relationship to their corresponding food chains and webs.

Energy and food production

Comparison of natural ecosystems and those based on modern intensive farming in terms of energy input and productivity.

Net productivity as defined by the expression

$$\text{Net productivity} = \text{Gross productivity} - \text{Respiratory loss}$$

The ways in which productivity is affected by farming practices that increase the efficiency of energy conversion. These include

- the use of natural and artificial fertilisers
- the use of chemical pesticides, biological agents and integrated systems in controlling pests on agricultural crops
- intensive rearing of domestic livestock.

Candidates should be able to

- apply their understanding of biological principles to present scientific arguments that explain how these and other farming practices affect productivity
 - evaluate economic and environmental issues involved with farming practices that increase productivity
 - consider ethical issues arising from enhancement of productivity.
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